

Léo Bougault

Nationality: French, Italian **Date of birth:** 29/06/1999 **Gender:** Male ✉ **Email address:**

leobougault@hotmail.fr  **LinkedIn:** <https://www.linkedin.com/in/l%C3%A9o-bougault-aa89b7160/> 

Address: Rue Clovis, 51100, Reims, France (Home)

ABOUT MYSELF

GNC Engineer (algorithms, flight simulator, HIL, CI/CD) at Latitude, graduate of Politecnico di Milano and ISAE-ENSMA.

WORK EXPERIENCE

Latitude - France, Reims

GNC Engineer

[01/12/2024 - Current]

- Development and validation of 6-DOF flight simulator (including actuators, sensors and OBC models)
- Development and testing of GNC algorithms into flight software
- Creation of HIL platform including C++ generated code, TwinCat real time integration and GNSS simulator
- Monte Carlo analysis for global system performance
- CI/CD development and automated V&V for GNC system

Skills: Matlab, Simulink, EcosimPro, STK, TwinCat, Embedded Coder

Latitude - France, Reims

Engineer Intern

[01/02/2024 - 31/08/2024]

- 0D turbopump model development
- Convex optimisation and CFD verification
- Validation through test campaign (post-process, fitting)

Skills: python (SciPy-SLSQP), git, ANSYS Fluent, LaTeX

ESA Academy - Belgium, Redu

Link: <https://arxiv.org/pdf/2502.13737>

System Engineer Intern

[01/04/2024 - 14/04/2024]

- 2024 Concurrent Engineering Challenge: Phase A mission development: launch injection, interplanetary transfer, aerobraking, transfer to target orbit, orbit propagation around Venus, station keeping and corrections
- 2023 Satellite design for Earth Observation: SAR system design

Skills: STK, ESA Comet

CNES - France, Paris

Propulsion Engineer Intern

[01/02/2022 - 30/06/2022]

- Rotodynamics and bearing mechanics for cryogenic application

- Multi-physics optimisation on Prometheus Turbopump

Skills: Python, RotorINSA, bearing CNES tool, Abaqus

Cranfield Univeristy - United Kingdom, Cranfield

Research Engineer Intern

[01/06/2021 - 31/08/2021]

- Injector development: multi-jet injection, turbulent mixing, flame stabilisation, residence time control

Skills: CATIA CAD, ANSYS Fluent, LaTeX

EDUCATION & TRAINING

Space Engineering Master Degree

Politecnico di Milano [01/09/2022 - 30/09/2024]

City: Milan | **Country:** Italy

Double degree program. Relevant courses:

- Estimation theory
- Attitude control
- Spacecraft guidance and navigation
- Orbital dynamics
- Space Propulsion & Launch system

Semester Exchange in Space and Astronautical Engineering Master degree

Sapienza University of Rome [01/09/2021 - 31/01/2022]

City: Rome | **Country:** Italy

- Spaceflight mechanics
- Control systems
- Part of Sapienza Space Team: CanSat competition

Engineering University

ISAE-ENSMA [01/09/2019 - 30/09/2024]

City: Poitiers | **Country:** France

Top French engineering graduate school in aerospace, mechanics and energy.

LANGUAGE SKILLS

Mother tongue(s): Italian | French

English

LISTENING: C1 **READING:** C1 **WRITING:** C1

SPOKEN PRODUCTION: C1

SPOKEN INTERACTION: C1

Spanish

LISTENING: B2 **READING:** B2 **WRITING:** B2

SPOKEN PRODUCTION: B1

SPOKEN INTERACTION: B1

HOBBIES AND INTERESTS

Trekking and trail running

Cello music conservatory